

HASSAN ISMAIL

CS & AI Student | Network & Security Engineer

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SUMMARY

Computer Science & Artificial Intelligence student at South Valley National University, working across two tracks: competitive programming in C++ (ECPC/ICPC) and network security. Completed FortiOS Administrator training at NTI and two Python courses through a Microsoft-backed MCIT program — applied hands-on in two verified FortiGate projects.

EDUCATION

South Valley National University

Aug 2024 – Jul 2028

B.Sc. Computer Science & Artificial Intelligence (in progress)

Currently in the third year.

National Telecommunication Institute (NTI)

Aug – Sep 2026

FortiOS Administrator — Network Security

60-hour official Fortinet Academy track: firewall policy, HA, SD-WAN, IPsec VPN, DMZ segmentation — built toward the NSE4 FortiOS Administrator certification (exam NSE4_FGT_AD-7.6).

CERTIFICATIONS

- Python Programming Language — Intermediate Level, Ministry of Communications & IT (MCIT) Egypt — Oct 2025
- Introduction to Programming Using Python, Ministry of Communications & IT (MCIT) Egypt — Sep 2025

SKILLS

Network Security: Firewall Policy Design, HA Clustering, Site-to-Site IPsec VPN, SD-WAN, Network Segmentation, CLI Diagnostics

Languages: C++, Python, HTML, CSS

Tools: FortiOS / FortiGate, Cisco Packet Tracer, EVE-NG

COMPETITIVE PROGRAMMING (ECPC / ICPC)

- ECPC Qualifications, Collegiate Programming Contest Day 8 — 94th Place (2025) → 93rd Place (2026), team
- ICPC ACPC Kickoff Online Individual Contest — Honorable Mention (2025), individual

APPLIED PROJECT EXAMPLE

Firewall Policy Engine — C++, personal project

- Built a rule-based firewall engine from scratch in C++: loads ACL-style policy from a file, evaluates packets against source/destination subnet (CIDR bitmask matching), protocol, and port range.
- Implements first-match-wins policy evaluation with implicit deny, plus a per-rule trace mode reporting exactly which condition(s) caused a match or rejection.

HA Cluster ↔ SD-WAN Site-to-Site VPN — FortiGate

- Built and connected two independent FortiGate deployments: an active-passive HA pair (Site P1) and a dual-WAN SD-WAN edge (Site P2), joined by a site-to-site IPsec tunnel with failsafe (blackhole) routing.
- Verified bidirectional reachability between sites via CLI-sourced pings — 0% packet loss.

DMZ Segmentation — FortiGate

- Designed a three-zone architecture (WAN / LAN / DMZ) on a single firewall to isolate public-facing services from the internal network.
- Enforced isolation via firewall policy: DMZ→LAN denied, LAN→DMZ and LAN→WAN permitted, implicit deny closing the rest.